Empirical Project
Machine Learning in Econometrics

”Double Machine Learning for Urban Climate Policy Evaluation: Congestion Pricing and Low-Emission Zones”

Department of Statistics
Ludwig-Maximilians-Universität München

Lukas Stank

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Supervised by Prof. Daniel Wilhelm

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1 Introduction

Human-induced climate change remains a pressing problem. This is best illustrated by the shrinking remaining carbon budget: at the start of 2026, only about 130 $GtCO_2$ could still be emitted without breaching safe warming limits [1], roughly three years of current emissions. Therefore, reducing emissions from major sectors is an urgent policy priority. Road transport is a major and policy-relevant source of these emissions, accounting for slightly more than 6 $GtCO_2$ in 2024, around 16% of global energy-related CO_2 emissions [2, 3]. Beyond the environmental damage, climate-related extreme events also impose large economic costs, illustrated by an estimated 822 billion euros in EU losses between 1980 and 2024, more than a quarter of which occurred in just the last four years of that period [4]. Reducing road-transport emissions can help limit both future environmental damage and its economic costs.

This report examines two urban policies intended to address these emissions: congestion pricing (CP), which charges drivers for entering within a defined urban area to reduce journeys and encourage public transport use, and low-emission zones (LEZ), which restrict access for vehicles that do not meet emission standards to encourage fleet renewal.

However, the effectiveness of both policies remains uncertain. Congestion pricing reduces traffic and local pollutants in some cities, for example modelling for London suggests CO_2 reductions of roughly 20% within the charging zone [5]. Low-emission zones improve air quality in several German cities [6], but since they target local pollutants rather than CO_2, they can shift vehicle purchases without reducing fleet-level CO_2. Consistent with this, an evaluation of the Madrid LEZ finds that the policy shifted purchases towards newer vehicles while average fleet CO_2 did not decline significantly [7]. Many studies also do not account for confounding factors [8], so observed changes may reflect other developments rather than the policy itself, making causal conclusions difficult.

Against this background, this report uses annual observations for 183 anonymised cities between 2008 and 2024 to estimate the causal effects of congestion pricing and low-emission zones on road-transport CO_2 emissions, addressing the following questions:

Q1: Were congestion pricing and low-emission zones effective in reducing road-transport CO_2 emissions? This provides a direct test of whether the two policies are effective climate-policy instruments.

Q2: Which types of cities benefit the most or least from these policies? Effects may vary with different city characteristics, such as population, income, density, public transport quality, or fuel prices, which shape how easily households substitute away from driving or replace vehicles. The answer can help cities with these characteristics assess whether the policies are likely to be particularly effective for them.

Q3: Does implementing both policies simultaneously produce a greater emission reduction than the sum of their individual effects? Congestion pricing reduces the number of journeys, while a low-emission zone changes the vehicles used for trips that remain. The two policies could therefore reinforce one another, making joint adoption more valuable for cities than either policy pursued alone.

2 Theory

2.1 Empirical strategy and identification

The panel covers $N = 183$ cities observed annually from 2008–2024, indexed by $i = 1, \dots, N$ (cities) and t (years) throughout. The outcome is $Y_{it} = \log(\text{transport\_co2}_{it})$, and D_{it} collects the policy indicators for congestion pricing and a low-emission zone (specified separately for each question below). We use the linear regression model

$$Y = D'\beta_1 + W'\beta_2 + \varepsilon, \qquad E(\varepsilon X) = 0, \qquad X = (D', W')', \tag{1}$$

where β_1 is the (low-dimensional) target parameter vector and β_2 the (potentially high-dimensional) nuisance parameter on W. W collects the available time-varying city characteristics, their pairwise interactions, and squared terms for a subset of key continuous controls. We exclude a few variables from W that are not suited to this role (see Section 3). Since cities may differ permanently in ways that drive both policy adoption and emissions, we remove a time-constant city effect from Y, D, and W via within-city demeaning (see Section 2.2). Model (1) applies to these demeaned variables.
For Q1 we use $D_{it} = (CP_{it}, LEZ_{it}, CP_{it} \times LEZ_{it})'$, so that β_{CP} and β_{LEZ} are the effects of each policy alone: $\beta_{CP} = 0$ ($\beta_{LEZ} = 0$) means no effect, $\beta_{CP} < 0$ ($\beta_{LEZ} < 0$) means congestion pricing reduces emissions, and $\beta_{CP} > 0$ ($\beta_{LEZ} > 0$) means it increases them. The effect of running both policies relative to neither follows as the point estimate $\beta_{CP} + \beta_{LEZ} + \beta_{\text{int}}$, with the same sign interpretation, or, given a separate estimate obtained directly from a regime-based specification, as reported in Appendix A.4.2.
For Q2 we extend D_{it} with policy $\times$ (centered) characteristic interactions, namely $CP_{it} \times (Z_{it} - \bar{Z})$ and $LEZ_{it} \times (Z_{it} - \bar{Z})$, with coefficient vectors $\beta_{CP\times Z}$ and $\beta_{LEZ\times Z}$: for a given characteristic Z_k, $\beta_{CP\times Z_k} = 0$ means the CP effect does not vary with Z_k, $\beta_{CP\times Z_k} < 0$ means cities with above-average Z_k benefit more from CP (a larger reduction) than the average city, and $\beta_{CP\times Z_k} > 0$ means they benefit less (or the effect reverses); analogously for $\beta_{LEZ\times Z_k}$.
For Q3, the same D_{it} as in Q1 also identifies β_{int}, which measures how far the joint effect of both policies deviates from simply summing β_{CP} and β_{LEZ}: $\beta_{\text{int}} = 0$ means the effects are simply additive, $\beta_{\text{int}} < 0$ indicates super-additive synergy, and $\beta_{\text{int}} > 0$ indicates sub-additive offsetting. That is, $\beta_{\text{int}} < 0$ means the joint effect reduces emissions more than the sum of the individual effects, while $\beta_{\text{int}} > 0$ means it reduces them less.

2.2 Proposed estimator

1. **Eliminate the city effect:** after specifying Y_{it}, D_{it}, and W_{it} as described above, we still need to address that cities may differ permanently in ways that drive both policy adoption and emissions, and these permanent differences would otherwise influence the estimated policy effect. To remove them, we replace Y_{it}, D_{it}, W_{it} by their within-city demeaned counterparts, reducing the panel model to (1) without a separate fixed-effect term. Appendix Section A.3 illustrates on a toy example why this within-transformation is necessary and sufficient for removing the city effect. Section 2.3 gives the formal justification.

2. **Nuisance learner:** for $E[Y \mid W]$ and $E[D \mid W]$ we use the `hdm` package's [9] `rlasso` function, a plug-in-penalty Lasso learner. Rather than choosing λ by cross-validating prediction error, this penalty is set by a closed-form formula calibrated to the regression's own noise level: a single Lasso fit per nuisance regression instead of an inner cross-validation loop.

3. **Estimation with `DoubleML`:** all remaining steps are carried out by the `DoubleML` package [10], which implements the cross-fitted partially linear regression (PLR) estimator of Chernozhukov et al. [11] for model (1):

   - *Cross-fitting:* the sample is split into $K = 5$ folds assigned at the city level, so a city's years are never split across training and evaluation, and the nuisance functions are estimated out-of-fold to avoid overfitting bias.
   - *Multiple treatments:* for a treatment vector with $q > 1$ columns, each coefficient $\beta_{1,l}$ comes from its own PLR run in which the remaining $q - 1$ treatment columns are added to that run's control side alongside W: the "one-by-one Double Lasso" procedure.
   - *Repeated cross-fitting:* since a single random fold split adds sampling variation of its own, the split is redrawn $n_{\text{rep}} = 5$ times and results are aggregated across splits to average out this variation.

2.3 Theoretical properties

Removing the city effect by within-city demeaning is exact and involves no estimation error. This is because the Frisch–Waugh–Lovell partialling-out theorem [12, Ch. 1] is applied twice: once to partial out the city means from Y and D, and once to partial out W from the demeaned Y and D. The first application is exact, since a city mean requires no estimation. The second is the standard Lasso-based projection, which does require estimation. See Appendix A.3 for the full argument and a worked numerical example.
The moment condition underlying $\hat{\beta}_1$ is Neyman orthogonal, at the true values of the nuisance parameters (γ_Y, γ_D), which represent the population projections of Y and D on W (proof in Appendix A.2). Orthogonality and cross-fitting address two distinct biases that arise from estimating these projections with a flexible ML method [12, Ch. 9]: orthogonality makes $\hat{\beta}_1$ locally insensitive to the nuisance estimate, so *regularization* bias does not propagate into $\hat{\beta}_1$. Cross-fitting evaluates the moment on data withheld from nuisance estimation, blocking *overfitting* bias. Neither suffices alone, but combined, they deliver an asymptotically unbiased and asymptotically normal estimator of β_1, with the required nuisance convergence rate satisfied by Lasso under approximate sparsity [12, Ch. 3].

3 Data preparation

Before estimation, we apply the following transformations to the data. The outcome is the log of transport emissions, $Y_{it} = \log(\text{transport\_co2}_{it})$, since the raw emissions distribution is strongly right-skewed. Taking logs of Y has a second benefit beyond fixing this skew: it turns every coefficient on D_{it} into a percentage effect on emissions rather than a raw change measured in kilotonnes. The exact percentage effect of a coefficient β on emissions is $100 \times (e^{\beta} - 1)$, so $\beta = 0$ ($e^{\beta} = 1$) means no effect, $\beta < 0$ ($e^{\beta} < 1$) means emissions shrink, and $\beta > 0$ ($e^{\beta} > 1$) means they increase. For example, $e^{\beta} = 0.8$ corresponds to a -20% effect, while $e^{\beta} = 1.2$ corresponds to a $+20\%$ effect. Due to skewness in several control variables as well, the untransformed variables population, population density, GDP per capita, city area, elevation, and tourism intensity enter in logs, and their raw levels are dropped to avoid duplicating the same information.
We build the baseline control set W by removing certain columns from the full data. We drop the outcome and its untransformed level (`transport_co2`), since it cannot be its own control. We drop the treatment indicators and the variables that define their timing (`cp_active`, `lez_active`, `cp_active`×`lez_active`, and the announcement and implementation year variables), since these are D itself. Local air quality (`pm25`) and the vehicle-fleet-composition shares (`fleet_diesel_share`, `fleet_electric_share`, `fleet_petrol_share`) are excluded as well, since these are downstream mechanisms of the policies rather than confounders. Total city-wide CO_2 (`total_co2`) is dropped too, since it mechanically contains the outcome, and so is `industry_public`, the omitted reference category, since the four industry shares sum to one. Everything that remains, including deliberately irrelevant variables such as museum visitors or bench counts, is a baseline control, which lets the Lasso's own variable selection serve as a built-in check that it discards variables with no real relationship to Y or D. From these baseline controls, W additionally includes squared terms for four key continuous variables (`log_population`, `log_gdp_pc`, `fuel_price`, `public_transit_score`) and all pairwise interactions among the baseline controls. After this processing, Y, D, and every column of W are within-city demeaned, as described in Section 2.2.

4 Empirical Evaluation

The following three subsections answer the politicians' three questions in turn, using the DML estimator and control set described above. In these plots, blue intervals indicate a significant emissions reduction, red would indicate a significant increase, and grey intervals include zero. Stars denote the estimate's significance level: one star (*) for $p < 0.05$, two stars (**) for $p < 0.01$, and three stars (***) for $p < 0.001$. Exact estimates for every coefficient discussed below are given in Appendix A.4.

4.1 Q1: Were the policies effective?

Figure 1 reports whether each policy reduces or increases average transport emissions, together with 95% confidence intervals. Congestion pricing comes out negative and significant: $\hat{\beta}_{CP} = -0.187$, corresponding to a -17.0% change in emissions. For the low-emission zone, by contrast, the coefficient is small and statistically insignificant: $\hat{\beta}_{LEZ} = -0.021$, or -2.1%.

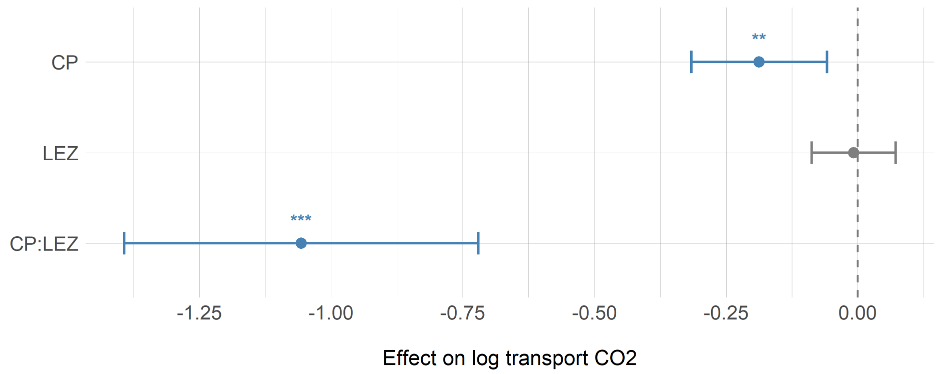

Figure 1: Average policy effects with city-clustered 95% confidence intervals. Exact estimates are reported in Appendix A.4.1.

The absence of a significant LEZ effect is consistent with the policy's design rather than a lack of statistical power. Low-emission zones restrict access to newer vehicles that meet stricter emission standards, yet these standards regulate local pollutants (NO_x, particulate matter), not CO_2. Since CO_2 emissions are directly proportional to fuel consumption, a newer vehicle that meets the LEZ standard emits substantially fewer pollutants, but roughly the same CO_2 per kilometre, as the older vehicle it replaces. The dominant behavioral response to an LEZ is thus fleet renewal rather than a reduction in driving, leaving total fuel consumption, and hence CO_2, essentially unchanged. This mechanism is supported by evidence from the Madrid LEZ, which increased registrations of alternative-fuel vehicles without significantly reducing the average CO_2 emissions of the newly registered cars [7]. Combining the two individual policy coefficients with the interaction term gives the estimated effect of implementing both policies simultaneously relative to a situation in which neither policy is active: $\hat{\beta}_{CP} + \hat{\beta}_{LEZ} + \hat{\beta}_{\text{int}} = -0.187 - 0.021 - 1.065 = -1.273$, which corresponds to a reduction in road-transport CO_2 emissions of approximately 72.0%. For valid confidence intervals on the effect of both policies vs. neither, see Appendix A.4.2.

4.2 Q2: Which cities benefit most?

Q2 asks how the policy effect varies across cities, i.e. the Conditional Average Treatment Effect, $\text{CATE}(z) := E[Y_{1,it} - Y_{0,it} \mid Z_{it} = z]$. We estimate this via its linear approximation, so that each interaction coefficient below gives the CATE's linear slope in one characteristic. We answer Q2 with a *primary specification* that interacts the treatments with nine pre-specified moderators, chosen before estimation by one criterion: Does the characteristic plausibly change how drivers and firms respond to the policy, rather than only being correlated with emissions levels or policy adoption? These nine cover four mechanisms: the availability of substitutes to driving (population density and public transit score directly, population and GDP per capita as scale/affordability proxies), a complementary price signal on driving (fuel price), traffic composition and exposure, since visitors and freight respond differently to local charges (tourism intensity, logistics activity, and coastal city), and public acceptance and enforcement (green party vote share). Candidates failing this criterion were deliberately left out, more directly: adoption-type variables that predict which cities choose policies and therefore belong in W as confounders, noisier or overlapping proxies for channels already captured by better measures (education share and unemployment, both proxying income or preferences; national climate pact and NGO environment index, both overlapping with green party vote share; and logistics industry share, overlapping with logistics activity), variables with no plausible channel to road-transport CO_2 (such as museum visitors or bench counts), as well as city area, which is mechanically impossible to include since density is the exact ratio of population and area. Interacting the treatments with every additional characteristic instead of only these nine would also come at a real cost: each extra interaction is another parameter the Lasso has to select and cross-fit around, which drastically increases both runtime and the difficulty of getting a stable estimate. A robustness check that does exactly this anyway, interacting the treatments with every baseline control with no discretion at all, is reported in Appendix A.4.5.

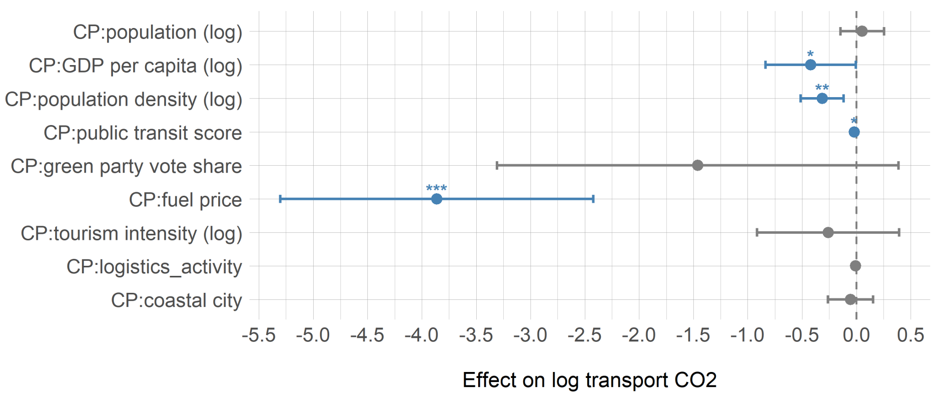

Figure 2: Heterogeneity of the CP effect. Exact estimates are reported in Appendix A.4.3.

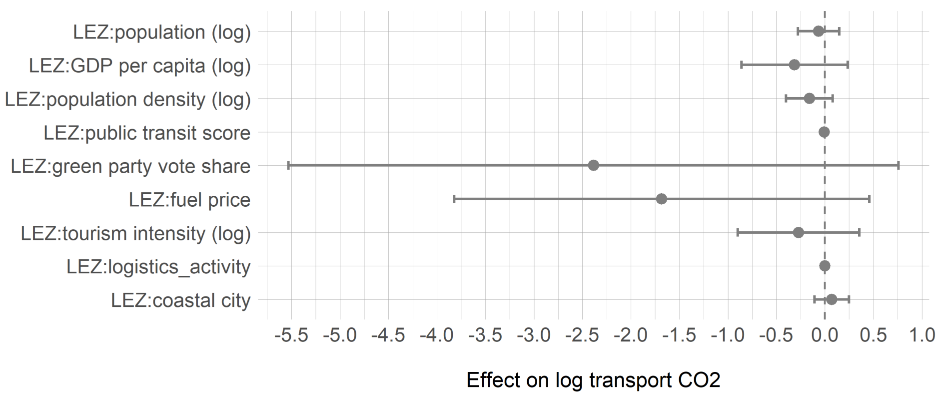

Figure 3: Heterogeneity of the LEZ effect. Exact estimates are reported in Appendix A.4.4.

Figure 2 shows the primary results for CP. Population density and the public transit score both come out significant with sensible signs: -0.337 for density, around -29%

per log-point of density above the sample mean, and −0.019 per point of transit score, around −1.9% per point. Congestion pricing cuts emissions more where giving up the car is more feasible, either because the city is denser or because it has a better transit score. Since these are two separate interaction terms rather than a joint one, our results do not tell us whether cities that are both dense and well-served by transit benefit even more, only that each characteristic matters on its own. The fuel-price interaction is also strongly significant, at −3.912. This intuitively makes sense: a higher fuel price already makes driving itself more expensive. In a city where fuel is expensive and a congestion charge is added on top, inhabitants therefore face two compounding costs of driving rather than one, which is a plausible reason for the CP effect to be more negative there. The magnitude, however, is implausibly large: in percentage terms it implies that a one-unit rise in fuel price would shift the CP effect by roughly 98%, letting differences of a few cents in fuel price swing the estimated effect far beyond what is plausible. Furthermore it collapses to −1.52 with only marginal significance in the full-specification robustness check (Appendix A.4), so we report it as suggestive at most. GDP per capita is only marginally significant here and loses significance entirely in that same check, so we do not feature it either. LEZ tells a simpler story: none of the nine moderators in Figure 3 reaches significance, in line with the insignificant average LEZ effect reported in Section 4.1. Taken together, congestion pricing pays off most in dense, transit-rich cities, while no city type in this sample shows a meaningful CO_2 response to the low-emission zone.

4.3 Q3: Do the policies reinforce each other?

The interaction coefficient in Figure 1 answers Q3 directly: $\hat{\beta}_{\text{int}} = -1.065$, significant with $p < 0.001$, corresponding to a −65.5% effect. The interaction is therefore strongly super-additive: the two policies together reduce emissions by substantially more than the sum of their individual effects. A plausible explanation is complementarity between the two policies, since congestion pricing reduces how much people drive while the LEZ changes the composition of vehicles on the road. When implemented together, the policies may therefore affect both the amount and the composition of road traffic in a way that neither policy achieves on its own. Combining the two individual coefficients with the interaction term gives an estimated total reduction of 72.0% relative to a situation in which neither policy is active. Since the standalone effects account for only a small part of this reduction, the combined effect is driven primarily by the interaction between the two policies.

4.4 Sensitivity analysis

We probe the main results along three dimensions. Each check changes exactly one thing relative to the baseline specification and uses the same DML estimator with repeated cross-fitting.

4.4.1 Nuisance learner

Table 1 replaces the baseline plug-in Lasso with cross-validated ridge and with a random forest. Ridge asks whether the result depends on the sparsity assumption specifically.

The forest checks whether the results change when the nuisance relationships are modelled more flexibly, since it can learn nonlinear patterns and interactions directly from the data. The ridge learner closely reproduces the baseline results. The CP and CP:LEZ estimates remain stable in sign, magnitude, and significance across all three learners, while the LEZ effect remains insignificant. The random forest, however, yields a significant negative LEZ coefficient of -0.272. This difference suggests that the LEZ estimate is sensitive to the choice of nuisance learner, should be interpreted cautiously, and would benefit from further study to understand the source of this sensitivity.

Table 1: Sensitivity to the nuisance learner.

Learner	Term	Estimate	SE	p	95% CI
rlasso (baseline)	CP	-0.187	0.070	0.008	$[-0.324, -0.049]$
	LEZ	-0.021	0.042	0.614	$[-0.104, +0.062]$
	CP:LEZ	-1.065	0.175	< 0.001	$[-1.408, -0.721]$
CV ridge	CP	-0.246	0.074	< 0.001	$[-0.390, -0.102]$
	LEZ	-0.048	0.055	0.385	$[-0.156, +0.060]$
	CP:LEZ	-1.122	0.155	< 0.001	$[-1.427, -0.817]$
random forest	CP	-0.406	0.049	< 0.001	$[-0.501, -0.310]$
	LEZ	-0.272	0.062	< 0.001	$[-0.394, -0.150]$
	CP:LEZ	-1.003	0.171	< 0.001	$[-1.338, -0.668]$

4.4.2 Control set W

In Table 2, we replace the baseline control set W with two alternative specifications. The *minimal* version keeps the main effect of every baseline control but drops all interactions and squared terms. The *extended* version adds back the mediator variables deliberately excluded in Section 3, namely local air quality (`pm25`) and the diesel and electric fleet shares. This specification therefore includes variables that may lie downstream of the policies rather than act as confounders. Notably, all three coefficients remain essentially unchanged in both alternative specifications. First, this suggests that the flexible interactions and squared terms in the baseline W contribute little beyond the main effects. Therefore, the *minimal* specification for W may be sufficient for the baseline analysis, substantially reducing computation time without changing the substantive conclusions. Second, the similarity of the estimates may indicate that our classification of `pm25` and the fleet shares as bad controls mattered less in practice than expected. However, this conclusion should be interpreted cautiously because the absence of a noticeable change does not imply that these variables are valid controls.

Table 2: Sensitivity to the control set W.

Control set	Term	Estimate	SE	p	95% CI
baseline	CP	-0.187	0.070	0.008	$[-0.324, -0.049]$
	LEZ	-0.021	0.042	0.614	$[-0.104, +0.062]$
	CP:LEZ	-1.065	0.175	< 0.001	$[-1.408, -0.721]$
minimal controls	CP	-0.199	0.063	0.002	$[-0.322, -0.075]$
	LEZ	-0.048	0.038	0.209	$[-0.122, +0.027]$
	CP:LEZ	-1.064	0.198	< 0.001	$[-1.452, -0.676]$
with added mediators	CP	-0.185	0.070	0.008	$[-0.322, -0.047]$
	LEZ	$+0.024$	0.047	0.609	$[-0.068, +0.116]$
	CP:LEZ	-0.964	0.173	< 0.001	$[-1.303, -0.625]$

4.4.3 Placebo treatment: announced but not yet implemented

The final check targets the identifying assumption itself rather than the estimator. A policy may become a major topic of local public discussion years before it is implemented. During this announcement window, the policy itself cannot yet have affected emissions, since no charge is collected and no vehicle is banned. The placebo coefficient comes from an indicator that switches on for the years after a policy is announced and off again once it is implemented. Conditional on the remaining controls and city fixed effects, it measures whether emissions are systematically higher or lower during this window than in other years for the same city. A coefficient close to zero indicates no systematic change before implementation. A significantly negative coefficient could possibly be explained by anticipatory reductions driven by public awareness of the policy, for example if discussion alone already changes household and firm behavior. A significantly positive coefficient could reflect a pre-existing upward trend that continues unaffected by the announcement, since nothing binding is yet in place, or endogenous timing, if cities tend to announce a policy precisely when their emissions are rising, or other developments coinciding with the announcement. Both readings are suggestive at best, and the positive case even more so than the negative one, since the test cannot distinguish between these explanations.
Table 3 shows the LEZ placebo is statistically insignificant. The CP placebo, however, is positive and significant at $+0.252$: emissions were higher during CP's announcement window than in other years. Since the coefficient alone cannot identify why, this raises the concern that the CP estimate may partly reflect developments already present before implementation, and should be interpreted with that caution in mind.

Table 3: Placebo check: announcement-window indicators, estimated jointly with the actual treatments.

Term	Estimate	SE	p	95% CI
CP	-0.143	0.076	0.060	$[-0.292, +0.006]$
LEZ	-0.010	0.044	0.829	$[-0.097, +0.077]$
CP:LEZ	-1.052	0.175	< 0.001	$[-1.395, -0.708]$
CP placebo	$+0.252$	0.094	0.007	$[+0.068, +0.437]$
LEZ placebo	$+0.107$	0.069	0.120	$[-0.028, +0.242]$

5 Discussion

Cross-fitted DML with within-city demeaning removes permanent city-level confounding and common shocks while still letting a rich, high-dimensional control set be selected flexibly. Taken together, our results paint a fairly clear picture of effectiveness: congestion pricing reliably reduces road-transport CO_2, with a significant -17.0% effect that holds up across alternative control sets and even strengthens under a more flexible random forest learner. The low-emission zone, by contrast, shows no significant average effect under our baseline specification, consistent with its focus on local pollutants rather than CO_2 directly. However, this null result is not fully robust, since the random forest specification instead yields a significant negative LEZ coefficient, so we do not rule out a CO_2 effect that our baseline learner is simply not flexible enough to detect. Most strikingly, the two policies together produce a strongly super-additive reduction of roughly 71–72%, a result confirmed by two independent approaches: summing the three separate coefficients, and directly estimating the combined effect from mutually exclusive policy-regime indicators (Appendix A.4.2). Still, the analysis has several limitations and possible extensions specific to this design:

1. **A full difference-in-differences specification would sharpen identification.** Our design only tests for a level effect once a policy is active; it cannot see what happens to emissions in the years just before implementation. The announcement placebo is a first attempt at this: it adds an indicator for the announcement-to-implementation window and checks whether its coefficient is zero. It comes out non-zero for CP, a warning sign whose possible causes (pre-trends, anticipatory adjustment, or non-linearity) are discussed in Section 4.4.3. A difference-in-differences (DiD) design [12, Ch. 16] would not resolve this problem, but would broaden our analysis with a more nuanced, year-by-year view of the period before adoption. We would estimate a separate coefficient for each year relative to each city's own adoption date rather than one pooled before/after number. Given the significant CP placebo coefficient (Section 4.4.3), we would expect at least some pre-adoption CP coefficients to be non-zero as well. A gradual pattern across several years would be more consistent with a pre-existing trend, while a sharp jump only in the year just before adoption would be more consistent with anticipatory reactions. Either way, our within-city demeaning cannot separate these from the policy's own effect, biasing the main estimates.

2. **Non-parametric CATE estimation**
   Q2 already estimates the CATE, but only as a linear approximation, imposing a functional form and raising a multiple-testing concern across the full set of interaction terms. A non-parametric approach would instead let the data determine the shape of the relationship, rather than imposing linearity. Possible implementations include fitting a regression tree, known as a causal tree [13], or a random forest, known as a causal forest [14]. Rather than separate coefficients per characteristic, such methods estimate the effect directly for any combination, such as a city that is both dense and coastal, with a valid confidence interval for that combined estimate.

A Appendix

A.1 Use of AI Tools

1. ChatGPT/Claude

- Code implementation and technical support: Assistance with translating algorithms from their mathematical formulation into R code, as well as solving practical issues in the implementation (e.g., plotting, parallelization, etc.).
- Text improvement and formulation assistance: Clarifying theoretical arguments, rephrasing sections for better readability, improving text flow and structure, stylistic refinement.
- Structure of the report: Support with organizing chapters, outlining sections, and improving the overall coherence of the work.
- Literature research: Support with identifying related work regarding emissions.

2. Quality Assurance

All content generated or improved by AI tools was critically reviewed and independently validated. In most topics above, AI tools offered a starting point that was then expanded upon. In the case of the codebase provided in the linked GitHub repository, the underlying structure was developed with the assistance of AI. However, not a single line of code was used without further inspection and supervision. The same applies to the few, singled-out text passages, which were partially generated with the assistance of AI. Final responsibility for all content lies with the author.

A.2 Neyman orthogonality of the score of model (1)

All variables below are understood to be the within-city demeaned quantities used in estimation, and D is understood to be a single column of the treatment vector at a time, with the remaining columns absorbed into W (the one-by-one procedure of Section 2.2). Model (1) states $Y = D'\beta_1 + W'\beta_2 + \varepsilon$: since W is potentially high-dimensional, we do not estimate β_2 directly, but instead identify β_1 by partialling W out of both Y and D.

Partialling out in the population: Let γ_Y, γ_D denote the population coefficient vectors

$$\gamma_Y \in \arg\min_{g\in\mathbb{R}^p} E\left[(Y - W'g)^2\right], \qquad \gamma_D \in \arg\min_{g\in\mathbb{R}^p} E\left[(D - W'g)^2\right],$$

and define the residualized variables

$$\tilde{Y} := Y - W'\gamma_Y, \qquad \tilde{D} := D - W'\gamma_D.$$

γ_Y, γ_D are nuisance parameters, not of direct interest, but needed to partial W out of Y and D before β_1 becomes identifiable. Their defining first-order conditions are the normal equations

$$E\left[W(Y - W'\gamma_Y)\right] = 0, \qquad E\left[W(D - W'\gamma_D)\right] = 0, \tag{2}$$

i.e. $E[W\tilde{Y}] = 0$ and $E[W\tilde{D}] = 0$. By the Frisch–Waugh–Lovell theorem,

$$\beta_1 = \frac{E[\tilde{D}\tilde{Y}]}{E[\tilde{D}^2]},$$

i.e. β_1 satisfies

$$E[\tilde{D}^2]\beta_1 = E[\tilde{D}\tilde{Y}] \iff E\left[\tilde{D}\tilde{Y} - \beta_1\tilde{D}^2\right] = 0 \iff E\left[(\tilde{Y} - \beta_1\tilde{D})\tilde{D}\right] = 0.$$

Deriving the moment equation: Substituting the definitions $\tilde{Y} = Y - W'\gamma_Y$ and $\tilde{D} = D - W'\gamma_D$ directly into this last equation gives

$$E\Bigg[\underbrace{\Big[\underbrace{Y - W'\gamma_Y}_{\tilde{Y}} - \beta_1\underbrace{(D - W'\gamma_D)}_{\tilde{D}}\Big]\underbrace{(D - W'\gamma_D)}_{\tilde{D}}}_{=:\,\psi(Y,D,W;\beta_1,\gamma)}\Bigg] = 0, \qquad \gamma = (\gamma_Y', \gamma_D')'.$$

More generally, define, for any candidate b_1 and any candidate $g = (g_Y', g_D')'$,

$$\psi(Y, D, W; b_1, g) := \left[Y - W'g_Y - b_1(D - W'g_D)\right](D - W'g_D), \tag{3}$$

so that the previous equation is exactly the moment condition

$$E\left[\psi(Y, D, W; \beta_1, \gamma)\right] = 0.$$

Plugging in the true values $b_1 = \beta_1$, $g = \gamma$ recovers the identifying equation above, while plugging in some other $g \neq \gamma$ (e.g. an estimate) generally does not give zero exactly. It is this sensitivity of $E[\psi(Y, D, W; \beta_1, g)]$ to g around $g = \gamma$ that we study next.
In the sample, γ_Y, γ_D are estimated by Lasso: $\hat{\gamma}_Y := \arg\min_{g\in\mathbb{R}^p} \mathbb{E}_n[(Y - W'g)^2] + \frac{\lambda}{n}\sum_{k=1}^{p}|g_k|$ and analogously for $\hat{\gamma}_D$, giving the sample residuals $\check{Y} = Y - W'\hat{\gamma}_Y$, $\check{D} = D - W'\hat{\gamma}_D$ and the estimator $\hat{\beta}_1 = \mathbb{E}_n[\check{D}\check{Y}]/\mathbb{E}_n[\check{D}^2]$

Proposition: The score in (3) is Neyman orthogonal with respect to the nuisance parameters at their true values:

$$\nabla_g E\left[\psi(Y, D, W; \beta_1, g)\right]\big|_{g=\gamma} = 0.$$

Proof: Since g_Y and g_D are finite-dimensional coefficient vectors, we differentiate the population moment with respect to them directly. With respect to g_Y,

$$\nabla_{g_Y} E\left[\psi(Y, D, W; b_1, g)\right] = -E\left[W(D - W'g_D)\right],$$

which is zero at $g_D = \gamma_D$ by the second equation in (2), for any b_1 and g_Y.
With respect to g_D, applying the product rule,

$$\nabla_{g_D} E\left[\psi(Y, D, W; b_1, g)\right] = 2b_1 E\left[W(D - W'g_D)\right] - E\left[W(Y - W'g_Y)\right].$$

Evaluated at $b_1 = \beta_1$, $g_Y = \gamma_Y$, $g_D = \gamma_D$, both terms vanish by (2), so this gradient is zero as well.
Since both partial gradients vanish at $g = \gamma$,

$$\nabla_g E\left[\psi(Y, D, W; \beta_1, g)\right]\big|_{g=\gamma} = 0,$$

so the score of model (1) is Neyman orthogonal.

A.3 Illustrative example: why within-city demeaning removes the city effect

The full model has three blocks of regressors: D (treatment), W (controls), and Ω (the full set of city dummies, never built explicitly, but conceptually present),

$$Y = D'\beta_1 + W'\beta_2 + \Omega'\alpha + \varepsilon.$$

Getting to β_1 from here takes two separate applications of FWL, not one:

- **FWL step 1:** partial Ω out of Y, D, and every column of W,

  $$\tilde{Y} = Y - BLP(Y \mid \Omega), \quad \tilde{D} = D - BLP(D \mid \Omega), \quad \tilde{W} = W - BLP(W \mid \Omega).$$

  Because Ω is a full set of mutually exclusive city dummies, $BLP(\cdot \mid \Omega)$ collapses to a simple group mean (shown below), so this step is just within-city demeaning, applied to Y, D, *and* W alike, and it is exact: no estimation error, since a population mean requires no statistical estimation at all.

- **FWL step 2:** partial the now-demeaned $\tilde{W}$ out of $\tilde{Y}$ and $\tilde{D}$, and regress the residuals on each other to get $\hat{\beta}_1$. Here $BLP(\cdot \mid \tilde{W})$ is *not* a simple mean anymore, since $\tilde{W}$ is high-dimensional and continuous, so this step is only approximate and must be estimated by Lasso. This is proven in Appendix A.2, whose opening line already treats Y, D, W as the within-city demeaned quantities from step 1.

This appendix works out step 1 in detail. Since step 1 acts on Y, D, and every column of W in exactly the same way, the illustrative example below keeps only Y and D for clarity. The identical demeaning applies to each control in W before step 2 ever begins.

Two equivalent ways to control for the city effect: Write the true model as $Y_{it} = c_i + \beta D_{it}$, where c_i is a permanent, city-specific effect (everything constant about city i over the sample period, observed or not) and β is the policy effect we want. There are two equivalent ways to keep c_i from contaminating the estimate of β, which yield numerically identical $\hat{\beta}$'s:

1. **Joint regression with city dummies:** let Ω be the full set of city dummy variables, $\Omega = (\mathbf{1}_{i=1}, \ldots, \mathbf{1}_{i=N})'$, and regress Y on D and Ω together,

$$Y = \beta D + \Omega'\gamma + \varepsilon.$$

   Each γ_i directly estimates c_i.

2. **The FWL route (demeaning, step 1 above):** partial Ω out of Y and D first, then regress the residuals on each other.

Why $BLP(Y \mid \Omega)$ is just the city mean when Ω is city dummies: By definition,

$$\gamma \in \arg\min_{g_1,\ldots,g_N} E\left[\left(Y - \textstyle\sum_{k=1}^{N} g_k \mathbf{1}_{i=k}\right)^2\right].$$

Because the dummies are mutually exclusive (every observation belongs to exactly one city), each observation's squared error depends on only one g_k, so this single minimization problem splits into N separate, unrelated problems, one per city:

$$\min_{g_i} E\left[(Y - g_i)^2 \mid \text{city} = i\right] \quad \text{for each } i.$$

The solution to each is the ordinary mean-squared-error minimizer, $g_i = E[Y \mid \text{city} = i]$, the mean of Y within that city. Hence

$$BLP(Y \mid \Omega) = \bar{Y}_i, \qquad BLP(D \mid \Omega) = \bar{D}_i,$$

and the FWL residuals $\tilde{Y} = Y - BLP(Y \mid \Omega)$, $\tilde{D} = D - BLP(D \mid \Omega)$ are exactly $Y_{it} - \bar{Y}_i$ and $D_{it} - \bar{D}_i$, yielding within-city demeaning. Nothing in this argument used any property of Y specifically, meaning the same collapse to a group mean holds for D and for every column of W, which is why step 1 above demeans all three alike. The following example makes this concrete and shows numerically why skipping this step (route 1 and route 2 both omitted, i.e. no city control at all) breaks the estimate.

Setup: Consider two cities, A and B, observed for two years, $t = 1$ (before) and $t = 2$ (after). City A has a permanently high baseline emissions level (e.g. it is an industrial hub) and adopts the policy in year 2. City B never adopts and never had that same permanent high baseline. Suppose the data are generated exactly by $Y_{it} = c_i + \beta D_{it}$, with true city effects $c_A = 10$, $c_B = 4$, and true policy effect $\beta = -1$:

	D_{it}	Y_{it}
City A, $t = 1$	0	10
City A, $t = 2$	1	9
City B, $t = 1$	0	4
City B, $t = 2$	0	4

The naive comparison suffers from omitted-variable bias: A pooled regression of Y on D (no city control at all, i.e. neither route above) estimates $E[Y \mid D=1] - E[Y \mid D=0] = 9 - \frac{10+4+4}{3} = 3$: it says the policy raised emissions, the opposite of the truth. Decomposing this comparison using the true data-generating process shows exactly where the error comes from:

$$\underbrace{E[Y \mid D=1] - E[Y \mid D=0]}_{=3} = (c_A + \beta) - \left(\tfrac{1}{3}c_A + \tfrac{2}{3}c_B\right) = \underbrace{\beta}_{=-1} + \tfrac{2}{3}\underbrace{(c_A - c_B)}_{=4},$$

i.e. the naive comparison equals the true effect plus a bias term equal to two-thirds of the permanent level gap between the two cities. The bias term is nonzero precisely because D is correlated with city identity here, since the only city that ever switches D on is also the one with the permanently higher baseline.

Demeaning removes the bias exactly: subtract each city's own mean, $\bar{Y}_A = 9.5$, $\bar{Y}_B = 4$, $\bar{D}_A = 0.5$, $\bar{D}_B = 0$:

	$D_{it} - \bar{D}_i$	$Y_{it} - \bar{Y}_i$
City A, $t=1$	-0.5	0.5
City A, $t=2$	0.5	-0.5
City B, $t=1$	0	0
City B, $t=2$	0	0

Because $Y_{it} - \bar{Y}_i = (c_i + \beta D_{it}) - (c_i + \beta \bar{D}_i) = \beta(D_{it} - \bar{D}_i)$, the city effect c_i cancels exactly, for any c_i. Therefore demeaning does not just reduce the omitted-variable bias derived above, it removes it completely, regardless of how large the permanent gap between A and B is. Regressing the demeaned Y on the demeaned D gives

$$\hat{\beta} = \frac{\sum_{i,t}(D_{it} - \bar{D}_i)(Y_{it} - \bar{Y}_i)}{\sum_{i,t}(D_{it} - \bar{D}_i)^2} = \frac{(-0.5)(0.5) + (0.5)(-0.5) + 0 + 0}{(-0.5)^2 + (0.5)^2 + 0 + 0} = \frac{-0.5}{0.5} = -1,$$

recovering the true $\beta = -1$ exactly. City B, which never varies in D, contributes $(0,0)$ in both rows and drops out entirely: it carries no information about the effect of switching the policy on, since it never switches. Only city A's within-city before/after comparison remains, uncontaminated by the level difference to city B. Once Y, D, and W have all been demeaned this way, Appendix A.2 carries out step 2: partialling the demeaned W out of the demeaned Y and D via Lasso to identify β_1.

A.4 Additional tables: exact estimates

A.4.1 Average policy effects and interaction (Figure 1)

Table 4: Average policy effects. Exact estimates underlying Figure 1.

Term	Estimate	SE	p	95% CI	Effect in %
CP	-0.187	0.070	0.008	$[-0.324, -0.049]$	-17.0
LEZ	-0.021	0.042	0.614	$[-0.104, +0.062]$	-2.1
CP:LEZ	-1.065	0.175	< 0.001	$[-1.408, -0.721]$	-65.5

The CP:LEZ row above is the same interaction effect discussed in Section 4.3.

A.4.2 Policy regimes: valid confidence intervals for the combined effect

The -72.0% combined effect reported in Section 4.1 is obtained by summing the CP, LEZ, and interaction coefficients. This sum has no directly valid confidence interval, since computing one would require the full covariance matrix across the three coefficients rather than just their individual standard errors. To obtain a valid confidence interval, Table 5 instead reports an alternative specification that regresses emissions directly on indicators for each policy regime (CP only, LEZ only, or both, relative to neither), so that the effect of implementing both policies is estimated as a single coefficient with its own standard error and confidence interval. The regime-based estimate is $\hat{\beta} = -1.240$, which corresponds to a -71.1% reduction in road-transport CO_2 emissions for cities with both policies active. This closely matches the -1.273 obtained by summing the three separate coefficients, corresponding to a -72.0% reduction. The small remaining difference is attributable to sampling variation across the two DML runs rather than to the omitted covariance terms.

Table 5: Effects by policy regime (CP only, LEZ only, or both, relative to neither city).

Regime	Estimate	SE	p	95% CI
CP only	-0.163	0.070	0.021	$[-0.301, -0.025]$
LEZ only	$+0.012$	0.041	0.764	$[-0.069, +0.094]$
CP & LEZ (both)	-1.240	0.142	< 0.001	$[-1.519, -0.962]$

A.4.3 Heterogeneity of the CP effect, primary specification (Figure 2)

Table 6: Heterogeneity of the CP effect, primary specification. Exact estimates underlying Figure 2. Bold rows are significant at $p < 0.05$.

Characteristic	Estimate	SE	p
Population (log)	0.054	0.103	0.599
GDP per capita (log)	**−0.420**	**0.212**	**0.047**
Population density (log)	**−0.315**	**0.100**	**0.002**
Public transit score	**−0.018**	**0.009**	**0.039**
Green party vote share	−1.460	0.942	0.121
Fuel price	**−3.862**	**0.735**	**< 0.001**
Tourism intensity (log)	−0.260	0.333	0.435
Logistics activity	−0.007	0.004	0.115
Coastal city	−0.053	0.106	0.615

A.4.4 Heterogeneity of the LEZ effect, primary specification (Figure 3)

Table 7: Heterogeneity of the LEZ effect, primary specification. Exact estimates underlying Figure 3.

Characteristic	Estimate	SE	p
Population (log)	−0.065	0.109	0.549
GDP per capita (log)	−0.313	0.280	0.262
Population density (log)	−0.159	0.123	0.194
Public transit score	−0.005	0.009	0.561
Green party vote share	−2.386	1.605	0.137
Fuel price	−1.683	1.092	0.123
Tourism intensity (log)	−0.272	0.319	0.394
Logistics activity	−0.001	0.004	0.828
Coastal city	0.071	0.091	0.434

A.4.5 Supplement: full heterogeneity specification (robustness check)

As a robustness check on Section 4.2, this specification interacts CP and LEZ with *every* baseline control, with no discretion in which characteristics are chosen. The density and public transit interactions from the primary specification survive here (Table 8): −0.341 for density and −0.019 for transit score, both still significant. A handful of additional CP interactions turn significant that were not among the nine pre-specified moderators (electricity price, renewable electricity share, national climate pact). These were left out of the primary specification for the reasons given in Section 4.2, such as national climate pact being judged redundant with the already-included green party vote share, not because they implausibly relate to road CO_2. With 34 characteristics tested per policy, some hits by chance alone are to be expected, so we interpret these three cautiously rather than as confirmed heterogeneity.

Table 8: Heterogeneity of the CP effect, full specification. Bold rows are significant at $p < 0.05$.

Characteristic	Estimate	SE	p
Unemployment	0.031	0.091	0.730
Education share	0.018	0.029	0.532
Public transit score	**−0.019**	**0.009**	**0.034**
Road km per capita	−0.162	0.143	0.260
Tourism intensity	−0.072	0.100	0.473
Logistics activity	−0.006	0.004	0.158
Green party vote share	0.800	0.982	0.415
Fiscal capacity	−0.696	0.515	0.176
Electoral competitiveness	−0.497	0.614	0.418
NGO environment index	0.002	0.003	0.526
National climate pact	**−0.308**	**0.157**	**0.049**
Electricity price	**−0.004**	**0.001**	**0.001**
Fuel price	−1.519	0.895	0.090
Temperature anomaly	−0.109	0.088	0.215
Precipitation	−0.0004	0.0005	0.462
Heating degree days	−0.0002	0.0002	0.306
Renewable electricity share	**−0.026**	**0.011**	**0.016**
Manufacturing industry share	−0.329	0.544	0.545
Services industry share	−1.021	0.527	0.053
Logistics industry share	−1.015	0.700	0.147
Elevation (log)	0.0002	0.0004	0.496
Coastal city	−0.068	0.100	0.498
Latitude zone 2	−0.007	0.169	0.965
Latitude zone 3	−0.050	0.211	0.814
Museum visitors per capita	−0.065	0.038	0.087
Library count	0.049	0.091	0.586
Streetlight density	0.008	0.005	0.144
Fountain count	−0.038	0.029	0.193
Bench count per capita	−0.014	0.022	0.524
Flagpole count	0.033	0.027	0.223
Sister city count	0.020	0.020	0.303
Population (log)	0.011	0.126	0.929
Population density (log)	**−0.341**	**0.104**	**0.001**
GDP per capita (log)	−0.200	0.321	0.532

Table 9: Heterogeneity of the LEZ effect, full specification.

Characteristic	Estimate	SE	p
Unemployment	0.034	0.089	0.699
Education share	0.029	0.023	0.213
Public transit score	−0.011	0.011	0.318
Road km per capita	−0.099	0.106	0.347
Tourism intensity	−0.098	0.085	0.252
Logistics activity	0.001	0.004	0.847
Green party vote share	−0.743	1.578	0.638
Fiscal capacity	0.737	0.838	0.379
Electoral competitiveness	−0.299	0.778	0.701
NGO environment index	−0.005	0.004	0.247
National climate pact	−0.060	0.127	0.634
Electricity price	−0.003	0.002	0.075
Fuel price	−0.055	1.576	0.972
Temperature anomaly	−0.123	0.080	0.124
Precipitation	−0.0002	0.0006	0.692
Heating degree days	−0.0003	0.0002	0.150
Renewable electricity share	−0.013	0.008	0.113
Manufacturing industry share	−0.953	0.522	0.068
Services industry share	−0.297	0.551	0.589
Logistics industry share	−0.897	0.807	0.266
Elevation (log)	0.0002	0.0003	0.510
Coastal city	0.085	0.095	0.373
Latitude zone 2	0.135	0.130	0.301
Latitude zone 3	0.092	0.219	0.675
Museum visitors per capita	−0.069	0.058	0.235
Library count	0.039	0.106	0.712
Streetlight density	0.005	0.003	0.140
Fountain count	−0.036	0.025	0.155
Bench count per capita	−0.022	0.026	0.387
Flagpole count	0.005	0.040	0.911
Sister city count	0.025	0.019	0.177
Population (log)	−0.134	0.152	0.376
Population density (log)	−0.158	0.106	0.135
GDP per capita (log)	−0.500	0.323	0.122

B Electronic appendix

Data, code and figures are provided in electronic form. Code is available on: `https://github.com/lukasstk/Empirical_project_ML_Econometrics.git`

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Declaration of authorship

"I confirm that this report is based on my own work. In preparing this report, I have not received any help from another human nor have I discussed any aspects of the empirical project with others."

Location, date

Name